Tian Yueyao

Curriculum Vitae

CONTACT

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Education Background

Bachelor: Software Engineering

Master: Signal Processing and Machine Learning

Research Interests

My research focuses on developing robust speaker recognition systems by integrating chaos theory with deep learning. Specifically, I am interested in:

- **Chaotic dynamical systems for audio feature extraction**
- **Deep learning architectures for speaker identification**
- **Robust speech processing under noisy conditions and short utterances**
- **Phase space reconstruction and nonlinear dynamics in speech signals**

Bachelor

University
2020 - 2024

Tiangong University, Tianjin, China
Major: Software Engineering
GPA: 3.52/4

Relevant Coursework

- Advanced Programming Language,
- Object-oriented Programming (JAVA),
- Data Structure, Advanced Mathematics,
- The Design and Analysis of Algorithm,
- Oracle Database Application and Development, Data Statistics and Analysis based on R Language, etc.

Master

University Sept 2024 - February 2026 (expected)

Tampere University, Finland
Major: Signal Processing and Machine Learning
GPA: 3.81/5

Relevant Coursework

- Advanced Signal Processing, Numerical Analysis
- Machine Learning, Deep learning
- Image and Video Processing, Audio and Speech Processing,
- Speech Processing
- Computational Imaging and Digital Holography, etc.

Technical Skills

- Programming Languages: Python, MATLAB, C, R
- Deep Learning Frameworks: PyTorch, TensorFlow
- Audio Processing: librosa, soundfile, torchaudio, Praat
- Tools & Platforms: Git, Linux, SLURM, LaTeX
- Specialized Skills: Chaos theory, speaker recognition, audio feature extraction, time-frequency analysis

Experience

Master's Research Projects June 2025 - Present
Chaotic Speaker Recognition System | *Master's Thesis Research*
Supervisor: Annamaria Mesaros

- Developing C-HiLAP (Chaotic Hierarchical Attractor Propagation) framework for speaker recognition
- Implementing chaotic dynamical systems (Lorenz, Rössler, Mackey-Glass) for audio feature extraction
- Applying phase space reconstruction using Takens' embedding theorem and RQA/MLSA analysis
- Conducting ablation studies on chaos synchronization loss
- Achieving 96% accuracy on LibriSpeech dataset

Experience

CNN-Based Automatic Music Transcription System
Tampere University | Advanced Audio Processing Course | Sept - Dec 2024

- Implemented multi-label classification system for piano/flute note detection using MAESTRO dataset
- Designed CNN architecture for audio-to-MIDI conversion with spectral feature extraction
- Analyzed precision-recall tradeoffs and temporal dependency limitations in note onset detection
- Evaluated model performance using frame-level and note-level metrics
- *Technologies: PyTorch, librosa, STFT, mel-spectrograms, MIDI processing*

Semantic Image Inpainting with GANs | Advanced Image Processing Course
Tampere University | Sept - Dec 2024

- Implemented GAN-based semantic inpainting system with context and prior loss optimization
- Achieved seamless integration of synthesized regions using Poisson blending
- Analyzed discriminator feedback's impact on visual realism for large missing regions
- *Technologies: PyTorch, GANs, latent space optimization, Poisson blending*

Undergraduate Research & Projects (2020-2024)
Robot Scientific Research | Research Team Leader
Tiangong University | Nov 2021 - Feb 2023

- Led interdisciplinary robotics research team focusing on obstacle avoidance algorithms and control systems
- Participated in research projects combining robotics with computer vision and motion planning
- Gained experience in cross-disciplinary collaboration and project management
- *Technologies: Robotics algorithms, control systems, computer vision*